Nishal Stanislaus Silva, *PhD*

Machine Learning Research Engineer | Predictive Modeling, Anomaly Detection, Time-Series | Applied ML

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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19th October 2026

Re: Senior Applied Machine Learning Scientist (predictive analytics) – TD

Dear Hiring Manager,

I am applying for the Senior Applied Machine Learning Scientist (predictive analytics) role at TD. I am a Machine Learning Engineer with a PhD (University of Trento, 2025) and 10+ years across industry and academia, and I build end-to-end ML systems that run in production under hard constraints. Predictive AI for regression, classification and anomaly detection across personal banking is the kind of work I want to lead, and I would bring a strong applied and research foundation to it.

Three parts of my work match your posting. For anomaly detection and structured data, at Forestpin I designed and deployed ML and statistical pipelines for financial forensics and compliance monitoring on large enterprise datasets, with SQL and Python scoring services for anomaly flagging and triage. For production ML, at MAS Holdings I led ML and computer vision systems that raised operational efficiency by up to 300%, and my published system runs at 14 ms on a Raspberry Pi 4, 74x faster than the DTW baseline, at F1 0.76 (Journal of the Audio Engineering Society, 2026). For evaluation and mentoring, I design benchmark suites with baselines and ablations, supervised two BSc theses at Trento, and mentored engineers and set up code review and CI/CD practice at MAS.

I should be direct about the gaps. I have not formally managed a team or owned performance reviews; my leadership has been technical lead work, mentoring and thesis supervision. I have not worked in banking or on credit and personal-lending models, and I have not used TD's Prism foundation models or worked with Layer 6. I have shipped the same pieces of the lifecycle, from data pipelines to monitoring and stakeholder requirements, and I expect to learn the domain and the governance and regulatory expectations quickly. The scale of TD's data and the chance to lead a team on predictive AI are what draw me to this role.

I am a Canadian permanent resident based in Toronto, ON, eligible to work without sponsorship, available immediately, and comfortable with the onsite arrangement in Toronto. I would welcome the opportunity to discuss how my experience can support the Predictive AI team at TD. My portfolio, publications and open datasets are available at nishal.xyz.

Sincerely,

Nishal Stanislaus Silva, PhD